

TARAsense Phase 1 – Consumer Test Analytics Module

This document outlines the recommended structure, workflow, and functional requirements for the TARAsense Consumer Test Analytics Module (Phase 1).

The goal is to guide developers in implementing statistically correct and user-friendly sensory analytics for hedonic scale consumer testing.

Consumer Test Dashboard

- |— Study Overview
- |— Per Sample Results
 - | |— Overall Liking
 - | |— Attribute Liking
 - | |— Penalty Analysis
 - | |— Mean Drop Analysis
 - | |— Graphs
- |
- |— Comparative Analysis
 - | |— Sample Selection
 - | |— Variable Selection
 - | |— Statistical Comparison
 - | |— Comparative Graph
- |
- |— Auto Interpretation

1. Main Objective

- Enable users to view consumer liking results per sample.
- Enable statistical comparison between samples.
- Automatically generate graphs and interpretations.
- Support penalty analysis and mean drop analysis.
- Keep statistical logic automated in the backend.

2. High-Level Workflow

- Raw Consumer Data
- Data Validation
- Sample-Based Summary
- Comparative Analysis
- Graph Generation
- Exportable Report

3. Study Overview Section

- Display study title
- number of consumers
- number of samples
- attributes evaluated
- hedonic scale used, and
- date conducted.

4. Sample-Based Results Section

- This section should present how each sample performed individually before comparison.

SAMPLE CARD VIEW

For each sample:

SAMPLE A

A. Overall Liking

Display:

- Mean liking score
- Standard deviation
- Interpretation

Example:

Metric	Value
Mean Overall Liking	7.4
Standard Deviation	1.2
Interpretation	Liked Moderately

4A. Overall Liking per Sample

- Display mean overall liking score.
- Display standard deviation.
- Display interpretation category.

4B. Attribute Liking per Sample

- Display mean and standard deviation for each sensory attribute.
- Example attributes: appearance, aroma, texture, sweetness, flavor, overall liking.

B. Attribute Liking

Display mean $\pm$ SD per attribute.

Example:

Attribute	Mean	SD
Appearance	7.2	1.1
Aroma	6.8	1.3
Texture	7.5	1.0
Sweetness	7.0	1.4

4C. Visualizations per Sample

- Automatically generate radar charts, bar graphs, and optional histograms.

4D. Penalty Analysis

- Display JAR percentage, Too Much percentage, Too Little percentage, Mean Drop, and Penalty Severity.

D. Penalty Analysis

Show:

- Attribute
- % JAR responses
- % Too Much
- % Too Little
- Mean Drop
- Penalty Severity

Example:

Attribute	JAR %	Too Much %	Too Little %	Mean Drop
Sweetness	45	40	15	1.3

Interpretation:

Sweetness is a major driver of disliking. 

4E. Mean Drop Analysis

- Display mean liking of JAR and non-JAR groups and corresponding mean drop.

E. Mean Drop Analysis

Display:

- Difference between JAR group and non-JAR group

Example:

Attribute	Mean Liking (JAR)	Mean Liking (Not JAR)	Mean Drop
Sweetness	7.8	6.2	1.6

5. Comparative Analysis Section

- Users should be able to compare selected samples and selected variables.

5A. Sample Selection Options

- Compare all samples.
- Compare selected samples only.

USER-CONTROLLED COMPARISON

The user should be able to choose:

Compare:

- ☐ All samples
- ☐ Sample A vs B
- ☐ Sample A vs C
- ☐ Custom selection

5B. Variable Selection

- Selectable variables include overall liking, appearance, aroma, texture, sweetness, penalty scores, and mean drop.

SELECT VARIABLE TO COMPARE

Dropdown:

Overall Liking
Appearance
Aroma
Texture
Sweetness
Penalty Scores
Mean Drop



COMPARISON OUTPUT

Example:

Sample	Mean Overall Liking
A	7.4
B	6.9
C	8.1

Statistics:

ANOVA p-value = 0.012
Result:
At least one sample differs significantly.



5C. Automatic Statistical Logic

- If two samples are selected, use t-test or Wilcoxon test.
- If three or more samples are selected, use ANOVA or Friedman test.
- Automatically generate p-values and determine significance.

5D. Comparative Output

- Display mean scores, standard deviations, p-values, and statistical interpretations.

5E. Automatic Graph Generation

- Generate bar charts with error bars, radar charts, penalty plots, and boxplots.

AUTO GRAPH GENERATION

Based on selected variable:

For overall liking:

- Bar chart with error bars

For attributes:

- Radar chart

For penalty:

- Penalty plot

For distribution:

- Boxplot

6. Automatic Interpretation Engine

- Generate plain-language insights automatically.
- Example: Sample C obtained the highest overall liking score.

7. Recommended Dashboard Structure

- Study Overview
- Per Sample Results
- Comparative Analysis
- Automatic Interpretation

8. Core Design Principle

- Users should not manually choose statistical tests.
- TARAsense should automatically determine the correct statistical analysis based on study structure and assumptions.

Key Principle: TARAsense should automate statistically correct sensory decision-making while maintaining a simple and visual user experience.

MOST IMPORTANT DESIGN PRINCIPLE

For Phase 1:

- Keep statistics intelligent in the backend
- Keep frontend simple and visual

The user should NOT need to choose:

- ANOVA
- Friedman
- Tukey
- Wilcoxon

TARAsense should decide automatically based on:

- number of samples
- study type
- data structure